

Quiz — 1.2.1 Operating Systems

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 · Component 01 · 1.2.1

Section A — Multiple choice (8 marks)

1. Which statement best describes virtual memory? A. Extra RAM chips added to increase memory capacity B. Using secondary storage as if it were RAM to run more/larger programs C. Cache memory built into the CPU D. A region of ROM reserved for the operating system Your answer: ☐
 2. Paging divides memory into blocks that are: A. Variable-sized and based on the program's logical structure B. Fixed-size C. The same size as the page file only D. Always 1 KB Your answer: ☐
 3. Which type of fragmentation is associated with segmentation? A. Internal fragmentation B. External fragmentation C. No fragmentation occurs D. Disk fragmentation only Your answer: ☐
 4. During interrupt handling, the contents of registers such as the PC and ACC are: A. Deleted to save memory B. Pushed onto a stack so the task can resume C. Written to the page file D. Sent to the printer buffer Your answer: ☐
 5. A real-time operating system (RTOS) is defined by its ability to: A. Run on many networked machines as one system B. Guarantee a response within a set time limit C. Allow many users to log in at once D. Fit entirely within ROM Your answer: ☐
 6. Which scheduling algorithm gives each process a fixed time slice (quantum) in turn? A. Shortest job first B. First come first served C. Round robin D. Shortest remaining time Your answer: ☐
 7. Which is the FIRST program to run when a computer is switched on? A. The operating system kernel B. The BIOS C. The device driver D. The loader Your answer: ☐
 8. Java bytecode run by the JVM is an example of: A. A system virtual machine running a whole guest OS B. Intermediate code run by a process virtual machine C. A device driver D. An interrupt service routine Your answer: ☐
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Section B — Short answer (12 marks)

9. State **two** differences between paging and segmentation. [2]

10. Explain what is meant by *disk thrashing* and why it slows a computer. [2]

11. Describe the steps the CPU/OS takes to handle an interrupt, making clear why the stack is needed.
[3]

12. A washing machine and a mainframe use different types of operating system. Name the most suitable OS type for each and justify your choice. **[2]**

13. Explain the role of a device driver and state why it is both hardware- and OS-specific. [3]

Section C — Exam-style question (5 marks)

14. A software company wants to test an unfamiliar piece of software on several different operating systems without buying extra physical computers. Explain how a **virtual machine** could be used to meet this need, and discuss **one** advantage and **one** drawback of this approach. [5]

[illegible]